MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 6

1. The basics

1. *For each of the following lists of vectors, determine if they are linearly independent. If they are not linearly independent (and thus they are linearly dependent) give a* ***nontrivial*** *dependence relation for the vectors:*

(1) $\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix}$

(2) $\begin{bmatrix}1 & 1\\0 & 0\end{bmatrix}, \begin{bmatrix}-1 & 0\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 2\\0 & 0\end{bmatrix}$

2. *Find an example of*

(1) three distinct linearly dependent vectors in $\mathbb{R}^4$;
(2) three linearly independent vectors in $\mathbb{R}^6$;
(3) three linearly dependent vectors in $\mathbb{R}^3$ such that each of these three vectors is not a scalar multiple of one of the other two vectors;
(4) three linearly dependent vectors in $\mathbb{R}^3$ such that any pair is of these three vectors is linearly independent.

3. *For each of the following matrices, find a minimal set of vectors* $\mathbf{a}_1, \ldots, \mathbf{a}_k$ *such that the column space of this matrix is equal to* $\text{span}\{\mathbf{a}_1, \ldots, \mathbf{a}_k\}$.

(1) $A = \begin{bmatrix}1 & -2 & 1\\2 & -4 & 1\\1 & -2 & 1\end{bmatrix}$

(2) $B = \begin{bmatrix}0 & 1 & 0\\1 & 0 & 0\\0 & 0 & 1\end{bmatrix}$

4. *Let* A *be an* 3×3*-matrix that has only one pivot position.*

(a) *Explain why there is a vector* $\mathbf{v} \in \mathbb{R}^3$ *such that* $\text{Col}(A) = \text{span}(\mathbf{v})$.

(b) *Is the same statement still true if* A *has two pivot positions?*

5. *Let* A *be an* $m \times n$*-matrix and* B *an* $n \times \ell$*-matrix. For each of the following statement, is it true or false? If true, explain why! If false, give a counterexample.*

(1) The column space of AB is contained in the column space of A.
(2) The null space of AB contained in the null space A.
(3) The set of all vectors $b \in \mathbb{R}^m$ that are not in the column space of A, is a subspace of $\mathbb{R}^m$.
(4) If the column space of A only contains the zero vector, then A is the zero matrix.

(5) The column space of A is equal to the column space $2A$.
(6) The column space of $A - I$ is equal to the column space of A.

6. *For each of the following statement, is it True or False? Give reasons!*

(1) The columns of a matrix A are linearly independent if the equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
(2) If S is a linearly dependent set, then each vector is a linear combination of the other vectors in S.
(3) The columns of any 4×5 matrix are linearly dependent.
(4) If $\mathbf{x}$ and $\mathbf{y}$ are linearly independent, and if $\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ is linearly dependent, then $\mathbf{z}$ is in $\text{span}\{\mathbf{x}, \mathbf{y}\}$.
(5) Two vectors are linearly dependent if and only if they lie on a line through the origin.
(6) If $\mathbf{x}$ and $\mathbf{y}$ are linearly independent, and if $\mathbf{z}$ is in $\text{span}\{\mathbf{x}, \mathbf{y}\}$, then $\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ is linearly independent.
(7) If a set in $\mathbb{R}^n$ is linearly dependent, then it contains at least $n + 1$ vectors.
(8) If $\mathbf{u}$ and $\mathbf{v}$ are linearly independent, then $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$ are linearly independent.

7. *Let $\mathbb{P}_2$ be the vector space of polynomials of degree at most 2.*

(1) Consider the following sets of polynomials in $\mathbb{P}_2$.

$$\mathcal{A} = \{1, t, t + 1\}$$
$$\mathcal{B} = \{1, t, t^2\}$$
$$\mathcal{C} = \{1, 2t, t^2, t^2 - t\}.$$

Which of these sets are linearly independent?
(2) Now consider $\mathcal{D} = \{t^2 + t + 1, t + 1, 1\}$. Is $\mathcal{D}$ linearly independent?
(3) Write $q(t) = t^2 + 1$ as a linear combination of the elements of $\mathcal{D}$.

8. *Consider the vector space V of functions from $\mathbb{R}$ to $\mathbb{R}$. Are x, $\sin(x)$, $\cos(x)$ linear independent vectors in this vector space? If so, explain why.*